

GenMol / UDLM engineering study

Complete V5 temperature screen and V6 Gibbs comparison | source snapshot 2026-09-07T10:09:32.675735+00:00

No superiority established. No formal promotion or final candidate lock. The highest observed repaired quality across the 18 engineering configurations is 57.03% (V6, s_t050_gibbs), versus 85.80% for the frozen local MDLM baseline. This descriptive maximum is selected from a small screen; it is not a confirmatory estimate.

Evidence	Configurations / runs / requested samples	Seed and compute budget
V5 temperature screen	12 configurations; 24 runs; 1,536 samples, all independently rescored	R/S/E x temperatures 0.50, 0.70, 0.85, 1.00; seeds 1200/1201; 128 predictor NFEs
V6 Gibbs comparison	6 configurations; 12 runs; 768 samples, all independently rescored	R/S/E x predictor/Gibbs; seeds 1300/1301; temperature 0.50; 128 total NFEs
Frozen local MDLM baseline	1 configuration; 3 runs; 3,000 samples	Seeds 0/1/2; released confidence sampling; inference and training compute not matched

128 samples per engineering configuration versus 3,000 baseline samples. Every engineering result averages two 64-sample seeds. All 2,304 V5+V6 requests remain disclosed; they are not pooled to estimate a single model configuration. V6 temperature 0.50 was chosen using partial V5 results, so it is an informed engineering choice, not a separately selected global optimum.

Training and checkpoints. Each R/S/E model starts from the same 50,000-update MDLM EMA and adds 1,000 updates at global batch 16 (16,000 example exposures; one GPU; microbatch 2; accumulation 8; seed 17). All three retain their original checkpoint bytes. The local MDLM training used batch 2,046 for 50,000 updates (102.3 million requested exposures); the paper used batch 2,048. A matched extra-update MDLM control is still absent. Scheduler and FiLM choices were selected on E loss, then shared across arms.

Arm definitions. R uses released uniform UDLM with its known loss/forward schedule mismatch; S repairs that schedule using uniform categorical diffusion; E uses the same repaired process and empirical SAFE prior with a 0.0002 uniform mixture. None is a published reproduction of a fully optimized UDLM molecular model.

Metrics and uncertainty. Validity = valid/requested; uniqueness = unique/valid; quality = distinct valid molecules with QED >= 0.6 and SA <= 4, divided by requested samples. QED summarizes drug-likeness; SA is the synthetic-accessibility score (lower is easier). Diversity is the released PyTDC fingerprint diversity on unique valid molecules. Means and sample SD average seeds equally; diversity is not pooled. Two-seed SD is fragile, not a confidence interval or significance test; configuration selection and molecular dependence add uncertainty.

Decoding and next work. Repaired metrics use SAFE fix=True and select the largest component by SMILES string length. Strict metrics use fix=False plus RDKit validation, without repair or component selection. Final UDLM seeds 0/1/2 remain reserved (the historical baseline already used those seeds). CE denoising with checkpoint-bound CE-to-LOO conversion is implemented and CPU-tested, but remains untrained and has no molecular results in this report. The archived V4 failure remains failed and contributes no ranked configuration.

Every configuration: mean +/- sample SD across two seeds

V/U/Q are percentages; D is diversity. P = predictor backbone evaluation; G = fresh Gibbs evaluation. All engineering rows use 128 samples/configuration and top-p=1.00. Baseline rows are context, not matched tests.

Study / arm	T	Kernel	Repair V%	Repair U%	Repair Q%	Repair D	Strict V%	Strict U%	Strict Q%	Strict D
V5 / R	0.50	128P	99.22 +/- 1.10	100.00 +/- 0.00	48.44 +/- 4.42	0.881 +/- 0.001	63.28 +/- 5.52	100.00 +/- 0.00	29.69 +/- 4.42	0.869 +/- 0.001
V5 / R	0.70	128P	97.66 +/- 3.31	100.00 +/- 0.00	44.53 +/- 1.10	0.894 +/- 0.002	46.09 +/- 1.10	100.00 +/- 0.00	19.53 +/- 5.52	0.875 +/- 0.002
V5 / R	0.85	128P	96.88 +/- 2.21	99.21 +/- 1.12	35.94 +/- 6.63	0.911 +/- 0.002	39.84 +/- 3.31	100.00 +/- 0.00	21.09 +/- 5.52	0.877 +/- 0.010
V5 / R	1.00	128P	93.75 +/- 0.00	99.17 +/- 1.18	30.47 +/- 1.10	0.911 +/- 0.005	37.50 +/- 6.63	100.00 +/- 0.00	14.06 +/- 2.21	0.881 +/- 0.003
V5 / S	0.50	128P	97.66 +/- 1.10	99.19 +/- 1.14	41.41 +/- 14.36	0.897 +/- 0.004	60.16 +/- 1.10	100.00 +/- 0.00	23.44 +/- 8.84	0.878 +/- 0.004
V5 / S	0.70	128P	97.66 +/- 1.10	100.00 +/- 0.00	42.19 +/- 19.89	0.898 +/- 0.006	42.97 +/- 1.10	100.00 +/- 0.00	21.09 +/- 1.10	0.864 +/- 0.004
V5 / S	0.85	128P	96.09 +/- 1.10	100.00 +/- 0.00	50.78 +/- 5.52	0.904 +/- 0.004	39.84 +/- 5.52	100.00 +/- 0.00	24.22 +/- 3.31	0.871 +/- 0.004
V5 / S	1.00	128P	92.97 +/- 7.73	98.41 +/- 2.24	32.03 +/- 1.10	0.924 +/- 0.005	25.00 +/- 0.00	100.00 +/- 0.00	11.72 +/- 1.10	0.886 +/- 0.002
V5 / E	0.50	128P	97.66 +/- 1.10	100.00 +/- 0.00	45.31 +/- 6.63	0.892 +/- 0.006	47.66 +/- 1.10	100.00 +/- 0.00	22.66 +/- 5.52	0.878 +/- 0.002
V5 / E	0.70	128P	98.44 +/- 0.00	99.21 +/- 1.12	45.31 +/- 11.05	0.899 +/- 0.003	40.62 +/- 11.05	100.00 +/- 0.00	21.09 +/- 9.94	0.872 +/- 0.015
V5 / E	0.85	128P	94.53 +/- 1.10	96.68 +/- 2.38	39.84 +/- 3.31	0.904 +/- 0.005	32.03 +/- 3.31	100.00 +/- 0.00	13.28 +/- 3.31	0.883 +/- 0.010
V5 / E	1.00	128P	93.75 +/- 4.42	99.14 +/- 1.22	35.94 +/- 6.63	0.920 +/- 0.012	17.97 +/- 3.31	100.00 +/- 0.00	9.38 +/- 4.42	0.884 +/- 0.007
V6 / R	0.50	128P	100.00 +/- 0.00	98.44 +/- 2.21	53.91 +/- 5.52	0.885 +/- 0.007	65.62 +/- 8.84	100.00 +/- 0.00	36.72 +/- 3.31	0.874 +/- 0.001
V6 / R	0.50	64P+64G	97.66 +/- 1.10	100.00 +/- 0.00	47.66 +/- 3.31	0.893 +/- 0.012	61.72 +/- 7.73	100.00 +/- 0.00	32.03 +/- 5.52	0.877 +/- 0.008
V6 / S	0.50	128P	98.44 +/- 2.21	99.19 +/- 1.14	54.69 +/- 4.42	0.882 +/- 0.006	61.72 +/- 1.10	100.00 +/- 0.00	33.59 +/- 1.10	0.866 +/- 0.007
V6 / S	0.50	64P+64G	98.44 +/- 2.21	97.61 +/- 1.18	57.03 +/- 1.10	0.892 +/- 0.005	55.47 +/- 7.73	100.00 +/- 0.00	36.72 +/- 1.10	0.878 +/- 0.004
V6 / E	0.50	128P	96.88 +/- 0.00	99.19 +/- 1.14	46.88 +/- 8.84	0.896 +/- 0.010	53.91 +/- 5.52	100.00 +/- 0.00	31.25 +/- 2.21	0.887 +/- 0.006
V6 / E	0.50	64P+64G	98.44 +/- 2.21	98.41 +/- 0.04	51.56 +/- 6.63	0.889 +/- 0.003	55.47 +/- 9.94	100.00 +/- 0.00	29.69 +/- 0.00	0.874 +/- 0.003
Local MDLM	0.50	released	100.00 +/- 0.00	99.87 +/- 0.06	85.80 +/- 1.23	0.823 +/- 0.001	98.80 +/- 0.20	99.87 +/- 0.06	84.77 +/- 1.22	0.822 +/- 0.002

The local MDLM row averages three 1,000-sample seeds. Published GenMol V1 repaired reference: validity 100.0%, uniqueness 99.7%, quality 84.6%, diversity 0.818 (paper run means). Hardware, training, data provenance and sample counts differ; higher diversity alone does not establish better molecules.

Within-arm V6 Gibbs deltas at fixed total NFE

Treatment minus predictor control, pairing seed 1300 with 1300 and 1301 with 1301 at the same checkpoint. Control: 128 predictor evaluations. Treatment: 64 predictor transitions + 64 fresh random-scan Gibbs updates. The paired RNG streams diverge because Gibbs changes draw counts. These are descriptive paired-seed deltas, not per-molecule paired effects or a confidence interval. Temperature is 0.50 in every pair.

Arm	Decoding	Delta validity (pp)	Delta uniqueness (pp)	Delta quality (pp)	Delta diversity
R	repaired	-2.34 +/- 1.10	+1.56 +/- 2.21	-6.25 +/- 2.21	+0.008 +/- 0.019
R	strict	-3.91 +/- 16.57	+0.00 +/- 0.00	-4.69 +/- 2.21	+0.003 +/- 0.009
S	repaired	+0.00 +/- 4.42	-1.59 +/- 2.32	+2.34 +/- 3.31	+0.010 +/- 0.001
S	strict	-6.25 +/- 6.63	+0.00 +/- 0.00	+3.12 +/- 0.00	+0.012 +/- 0.011
E	repaired	+1.56 +/- 2.21	-0.78 +/- 1.18	+4.69 +/- 2.21	-0.007 +/- 0.007
E	strict	+1.56 +/- 15.47	+0.00 +/- 0.00	-1.56 +/- 2.21	-0.013 +/- 0.003

The means show mixed effects: E improves repaired quality but loses strict quality; S improves both quality means while strict validity falls; R declines in both quality and validity branches. There is no universal Gibbs gain. V5 and V6 use different seeds, so their differences are not paired comparisons.

A random-scan Gibbs step resamples one uniformly chosen editable coordinate per sequence and may leave its value unchanged. Exact stationarity requires true compatible LOO conditionals and untempered sampling; these learned predictions at temperature 0.50 form an approximate corrector. This comparison also trades away half the predictor transitions, so it evaluates that allocation, not adding free extra compute.

Checkpoint / budget	SHA-256 (original source bytes; not reserialized)
MDLM 50k EMA warm-start source	8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6
R: +1,000 updates / +16,000 exposures	d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c
S: +1,000 updates / +16,000 exposures	100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72
E: +1,000 updates / +16,000 exposures	dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005

Artifact trail. The input manifest hashes both report JSON/CSV/PDF files, all 36 run summaries and raw CSVs, both protocols and their configuration bytes, the three continuation training summaries, the local baseline aggregate and its verified PDF, and this generator. Checkpoint identities are inherited from the independently validated generation/training records; this overview does not reload multi-gigabyte checkpoints.

Unmodified appendices follow: complete independently rescored V5 report; complete independently rescored V6 report; frozen local MDLM baseline report. They retain original page labels and historical context. The companion CSV/JSON contain all configuration means, SDs, paired deltas, hashes and original identifiers. No significance test, superiority declaration, formal promotion, or reserved-seed UDLM generation occurred here.

GenMol UDLM engineering exploration

engineering-v5-temperature | complete | 2026-09-07T09:50:06.376168+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

The local MDLM model completed 50,000 updates; each UDLM arm starts from its EMA weights and adds 1,000 updates. Training compute is not matched.

The scheduler and FiLM conditioner were selected on E denoising loss and then shared by R/S/E. This is not independent optimization of each method.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED ≥ 0.6 and SA ≤ 4 . Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdmlm_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_p100	2/2	0.9766 +/- 0.0110	1.0000 +/- 0.0000	0.4531 +/- 0.0663	0.8921 +/- 0.0055
e_t070_p100	2/2	0.9844 +/- 0.0000	0.9921 +/- 0.0112	0.4531 +/- 0.1105	0.8994 +/- 0.0028
e_t085_p100	2/2	0.9453 +/- 0.0110	0.9668 +/- 0.0238	0.3984 +/- 0.0331	0.9038 +/- 0.0054
e_t100_p100	2/2	0.9375 +/- 0.0442	0.9914 +/- 0.0122	0.3594 +/- 0.0663	0.9201 +/- 0.0115

Config	Seeds complete	validity	uniqueness	quality	diversity
s_t050_p100	2/2	0.9766 +/- 0.0110	0.9919 +/- 0.0114	0.4141 +/- 0.1436	0.8967 +/- 0.0043
s_t070_p100	2/2	0.9766 +/- 0.0110	1.0000 +/- 0.0000	0.4219 +/- 0.1989	0.8983 +/- 0.0064
s_t085_p100	2/2	0.9609 +/- 0.0110	1.0000 +/- 0.0000	0.5078 +/- 0.0552	0.9045 +/- 0.0044
s_t100_p100	2/2	0.9297 +/- 0.0773	0.9841 +/- 0.0224	0.3203 +/- 0.0110	0.9242 +/- 0.0053
r_t050_p100	2/2	0.9922 +/- 0.0110	1.0000 +/- 0.0000	0.4844 +/- 0.0442	0.8809 +/- 0.0010
r_t070_p100	2/2	0.9766 +/- 0.0331	1.0000 +/- 0.0000	0.4453 +/- 0.0110	0.8942 +/- 0.0019
r_t085_p100	2/2	0.9688 +/- 0.0221	0.9921 +/- 0.0112	0.3594 +/- 0.0663	0.9114 +/- 0.0020
r_t100_p100	2/2	0.9375 +/- 0.0000	0.9917 +/- 0.0118	0.3047 +/- 0.0110	0.9110 +/- 0.0051

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_p100	2/2	0.4766 +/- 0.0110	1.0000 +/- 0.0000	0.2266 +/- 0.0552	0.8785 +/- 0.0021
e_t070_p100	2/2	0.4062 +/- 0.1105	1.0000 +/- 0.0000	0.2109 +/- 0.0994	0.8718 +/- 0.0153
e_t085_p100	2/2	0.3203 +/- 0.0331	1.0000 +/- 0.0000	0.1328 +/- 0.0331	0.8833 +/- 0.0101
e_t100_p100	2/2	0.1797 +/- 0.0331	1.0000 +/- 0.0000	0.0938 +/- 0.0442	0.8839 +/- 0.0066
s_t050_p100	2/2	0.6016 +/- 0.0110	1.0000 +/- 0.0000	0.2344 +/- 0.0884	0.8781 +/- 0.0036
s_t070_p100	2/2	0.4297 +/- 0.0110	1.0000 +/- 0.0000	0.2109 +/- 0.0110	0.8641 +/- 0.0042
s_t085_p100	2/2	0.3984 +/- 0.0552	1.0000 +/- 0.0000	0.2422 +/- 0.0331	0.8713 +/- 0.0036
s_t100_p100	2/2	0.2500 +/- 0.0000	1.0000 +/- 0.0000	0.1172 +/- 0.0110	0.8859 +/- 0.0015
r_t050_p100	2/2	0.6328 +/- 0.0552	1.0000 +/- 0.0000	0.2969 +/- 0.0442	0.8686 +/- 0.0007
r_t070_p100	2/2	0.4609 +/- 0.0110	1.0000 +/- 0.0000	0.1953 +/- 0.0552	0.8749 +/- 0.0016
r_t085_p100	2/2	0.3984 +/- 0.0331	1.0000 +/- 0.0000	0.2109 +/- 0.0552	0.8772 +/- 0.0103
r_t100_p100	2/2	0.3750 +/- 0.0663	1.0000 +/- 0.0000	0.1406 +/- 0.0221	0.8811 +/- 0.0030

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
e_t050_p100	1200	completed	64	64	0.5000	0.2656	9.71
e_t050_p100	1201	completed	64	64	0.4062	0.1875	9.17
e_t070_p100	1200	completed	64	64	0.5312	0.2812	9.55
e_t070_p100	1201	completed	64	64	0.3750	0.1406	9.09
e_t085_p100	1200	completed	64	64	0.3750	0.1094	9.58
e_t085_p100	1201	completed	64	64	0.4219	0.1562	9.24
e_t100_p100	1200	completed	64	64	0.3125	0.0625	9.46
e_t100_p100	1201	completed	64	64	0.4062	0.1250	9.70
s_t050_p100	1200	completed	64	64	0.5156	0.2969	9.42
s_t050_p100	1201	completed	64	64	0.3125	0.1719	9.15
s_t070_p100	1200	completed	64	64	0.5625	0.2188	9.41
s_t070_p100	1201	completed	64	64	0.2812	0.2031	9.05
s_t085_p100	1200	completed	64	64	0.5469	0.2656	10.77
s_t085_p100	1201	completed	64	64	0.4688	0.2188	9.01
s_t100_p100	1200	completed	64	64	0.3281	0.1094	10.08
s_t100_p100	1201	completed	64	64	0.3125	0.1250	9.10
r_t050_p100	1200	completed	64	64	0.5156	0.3281	9.99
r_t050_p100	1201	completed	64	64	0.4531	0.2656	9.02
r_t070_p100	1200	completed	64	64	0.4531	0.2344	9.30
r_t070_p100	1201	completed	64	64	0.4375	0.1562	8.96
r_t085_p100	1200	completed	64	64	0.4062	0.2500	9.33
r_t085_p100	1201	completed	64	64	0.3125	0.1719	9.02
r_t100_p100	1200	completed	64	64	0.3125	0.1250	9.35
r_t100_p100	1201	completed	64	64	0.2969	0.1562	9.06

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={ 'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected { 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0' } - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: ecd4722c23debe193aeb50c034122ca20e501cf7c379e4beb6514b4d496d1a7

v5-e-t050-p100 / seed 1200: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3", "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t050-p100/seed_1200/raw_samples.csv", "sha256": "575549aadb838cc83426a13fe3bc3233922f0f4b8ca79a618a6ba630e6e9f414", "size_bytes": 20244, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}}

v5-e-t050-p100 / seed 1201: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5", "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t050-p100/seed_1201/raw_samples.csv", "sha256": "8894501102bb1dc32ae64feec0d3e932dd7a24a364ae4ca6dc74d76db8e7febb", "size_bytes": 20232, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}}

v5-e-t070-p100 / seed 1200: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3", "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t070-p100/seed_1200/raw_samples.csv", "sha256": "a0832e0b5489c32d25e9732b24e9160b32c48c485b39406f45d3a54c0ecc7cf8", "size_bytes": 20130}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t070-p100 / seed 1201: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5", "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t070-p100/seed_1201/raw_samples.csv", "sha256": "ab42728b2bea5adc25950a81f569a3df5c94b8f6a1ebdc8f585edd4631df541d", "size_bytes": 19412}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t085-p100 / seed 1200: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5", "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t085-p100/seed_1200/raw_samples.csv", "sha256": "06f0d2cfcf3215f70541d7c88c00cec89579b47d0975a8e99efdf35566240a8d", "size_bytes": 19118}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t085-p100 / seed 1201: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3", "raw": {"relative_path":

"output/udlm/engineering_v5/v5-e-t085-p100/seed_1201/raw_samples.csv", "sha256": "327e7b74d31709973085a72743b62661abe2663f0969ea8ae44237ee6b1d88de", "size_bytes": 19054}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t100-p100 / seed 1200: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t100-p100/seed_1200/raw_samples.csv", "sha256": "c5441a763ac9843d4a0865bbb0f33db9e47a9a1c55d2195de1bc29d1bbde9e51", "size_bytes": 18276}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t100-p100 / seed 1201: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t100-p100/seed_1201/raw_samples.csv", "sha256": "dd15322742447031c8d04ef61c37f6e0a98f46f32c57ff4e50a258b32ef20693", "size_bytes": 18294}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t050-p100 / seed 1200: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t050-p100/seed_1200/raw_samples.csv", "sha256": "67750d50a016eae3431aa95d3ec47a4976e1c6295323bc69777bcefa49d4caf9", "size_bytes": 21059}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t050-p100 / seed 1201: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy":

{ "active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": { "compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": "3", "raw": { "relative_path": "output/udlm/engineering_v5/v5-s-t050-p100/seed_1201/raw_samples.csv", "sha256": "1de893cf2f402a2021f8eb08dd6e9eca5a55e3627ada5dc54b990c1e7fe04af2", "size_bytes": 21005, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05" }

v5-s-t070-p100 / seed 1200: { "checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": { "diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7, "gpu": { "cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": { "compute_mode": "Default", "compute_processes": { "pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880, "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "launch_policy": { "active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": { "compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": "5", "raw": { "relative_path": "output/udlm/engineering_v5/v5-s-t070-p100/seed_1200/raw_samples.csv", "sha256": "ca74886bf6be3cee9a7a51f43724141fb63bd9ffd26b6ebc5a99c658d6cb8290", "size_bytes": 19956, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05" }

v5-s-t070-p100 / seed 1201: { "checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": { "diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7, "gpu": { "cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": { "compute_mode": "Default", "compute_processes": { "pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880, "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "launch_policy": { "active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": { "compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": "3", "raw": { "relative_path": "output/udlm/engineering_v5/v5-s-t070-p100/seed_1201/raw_samples.csv", "sha256": "fc3769c7bcf5071f1b8984559bb86f1abac89b8c07b47828c0770b8d2081b298", "size_bytes": 19596, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05" }

v5-s-t085-p100 / seed 1200: { "checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": { "diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85, "gpu": { "cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": { "compute_mode": "Default", "compute_processes": { "pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880, "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 5, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "launch_policy": { "active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": { "compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": "3", "raw": { "relative_path": "output/udlm/engineering_v5/v5-s-t085-p100/seed_1200/raw_samples.csv", "sha256": "db612b3231bc4f17e07b052863303215c6de0d5930e8a8114adced6f1ed89f7d", "size_bytes": 19991, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05" }

v5-s-t085-p100 / seed 1201: { "checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": { "diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0,

"softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t085-p100/seed_1201/raw_samples.csv", "sha256": "86c26ea024fb46143df0f5de9c67801411e5489c2a6db8dad7e37064c4e2b947", "size_bytes": 19489}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t100-p100 / seed 1200: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t100-p100/seed_1200/raw_samples.csv", "sha256": "6356b28ba7eeea9c2128ac0add101b0705c8f408b2309963798fa71fb282be8e", "size_bytes": 18533}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t100-p100 / seed 1201: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t100-p100/seed_1201/raw_samples.csv", "sha256": "f56974a534d8a39435858a77c3c341c459780fb6f4dda83798bd4ab55430ae34", "size_bytes": 18519}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t050-p100 / seed 1200: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t050-p100/seed_1200/raw_samples.csv", "sha256": "f941d5b4cc21edeef012814945ec2ac534d0d5ab655336e93ef4ac1b821168e6", "size_bytes": 21718}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t050-p100 / seed 1201: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 5}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t050-p100/seed_1201/raw_samples.csv", "sha256": "d27ee7308b3596eb46dae8c165aafa22ca96aeb21135a6411d7ab030c7ce2b28", "size_bytes": 20790}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t070-p100 / seed 1200: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 5}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t070-p100/seed_1200/raw_samples.csv", "sha256": "bbbeb58d2bbb6faeece43f13fc7a21d2204137f61406eac9a48e7f352d42cd820", "size_bytes": 20237}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t070-p100 / seed 1201: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 3}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t070-p100/seed_1201/raw_samples.csv", "sha256": "f9b92cd25a1cff261b49f516fc350bda523480b5320e59c8f2b259bcf3b0adc6", "size_bytes": 20340}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t085-p100 / seed 1200: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 5}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t085-p100/seed_1200/raw_samples.csv", "sha256": "81d10fb2e12e415f8c54fb1471ddb2a22242a90a9b18c2b8e672a0edc05ce72", "size_bytes": 19906}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t085-p100 / seed 1201: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t085-p100/seed_1201/raw_samples.csv", "sha256": "76cb87c29b6008868ad7a92ee99bc1d266030d4290f0b344185c8f99aa5f89b0", "size_bytes": 19645}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t100-p100 / seed 1200: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t100-p100/seed_1200/raw_samples.csv", "sha256": "a6a569384cc07e077edc60ce7e3883a0b4866a62d64024a6526dad5dea81dfa3", "size_bytes": 19414}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t100-p100 / seed 1201: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t100-p100/seed_1201/raw_samples.csv", "sha256": "a1bb88ced33cd8c2b6716b415fcfe2a83e54cd7d9eb2012d676be09ff54f3049", "size_bytes": 19966}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

GenMol UDLM engineering exploration

engineering-v6-gibbs | complete | 2026-09-07T10:09:32.675735+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

The local MDLM model completed 50,000 updates; each UDLM arm starts from its EMA weights and adds 1,000 updates. Training compute is not matched.

The scheduler and FiLM conditioner were selected on E denoising loss and then shared by R/S/E. This is not independent optimization of each method.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED ≥ 0.6 and SA ≤ 4 . Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Prospective sampling design

```
{"arms": ["R", "S", "E"], "claim_boundary": "Learned/tempered LOO conditionals are approximate; exact Gibbs stationarity is established only for the toy true-conditional tests.", "corrector_coordinate_selection": "one uniformly selected editable coordinate per row, fresh prediction at the post-predictor state and time", "frozen_before_v6_generation": true, "gibbs_treatment": {"corrector_updates": 64, "nfe": 128, "predictor_transitions": 64}, "partial_v5_observed_before_design": "E and S temperature outcomes through 0.85; no final evaluation seeds. Temperature 0.5 is fixed across arms to favor syntactic fidelity, not selected as the global quality winner.", "predictor_control": {"corrector_updates": 0, "nfe": 128, "predictor_transitions": 128}, "temperature": 0.5, "top_p": 1.0}
```

Exploratory parameter selection, not a confirmatory benchmark.

Only 1000 UDLM continuation updates: matched method training superiority is not isolated.

Historical baseline has 1000 samples per seed versus 64 here; diversity and uniqueness vary with sample count.

Strict molecular validity is reported separately from repaired validity.

The corrector changes transition allocation at fixed total NFE, not training.

Temperature 0.5 is a common engineering choice informed by partial v5 syntax diagnostics.

Corrector predictions can depend on their own noisy token; exact stationarity is not asserted for the learned model.

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdln_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
e_t050_predictor	1300	128	128	0
e_t050_predictor	1301	128	128	0
e_t050_gibbs	1300	128	64	64
e_t050_gibbs	1301	128	64	64
s_t050_predictor	1300	128	128	0
s_t050_predictor	1301	128	128	0
s_t050_gibbs	1300	128	64	64
s_t050_gibbs	1301	128	64	64
r_t050_predictor	1300	128	128	0
r_t050_predictor	1301	128	128	0
r_t050_gibbs	1300	128	64	64
r_t050_gibbs	1301	128	64	64

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_predictor	2/2	0.9688 +/- 0.0000	0.9919 +/- 0.0114	0.4688 +/- 0.0884	0.8959 +/- 0.0102
e_t050_gibbs	2/2	0.9844 +/- 0.0221	0.9841 +/- 0.0004	0.5156 +/- 0.0663	0.8886 +/- 0.0033

Config	Seeds complete	validity	uniqueness	quality	diversity
s_t050_predictor	2/2	0.9844 +/- 0.0221	0.9919 +/- 0.0114	0.5469 +/- 0.0442	0.8822 +/- 0.0064
s_t050_gibbs	2/2	0.9844 +/- 0.0221	0.9761 +/- 0.0118	0.5703 +/- 0.0110	0.8921 +/- 0.0052
r_t050_predictor	2/2	1.0000 +/- 0.0000	0.9844 +/- 0.0221	0.5391 +/- 0.0552	0.8850 +/- 0.0070
r_t050_gibbs	2/2	0.9766 +/- 0.0110	1.0000 +/- 0.0000	0.4766 +/- 0.0331	0.8932 +/- 0.0124

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_predictor	2/2	0.5391 +/- 0.0552	1.0000 +/- 0.0000	0.3125 +/- 0.0221	0.8874 +/- 0.0056
e_t050_gibbs	2/2	0.5547 +/- 0.0994	1.0000 +/- 0.0000	0.2969 +/- 0.0000	0.8744 +/- 0.0029
s_t050_predictor	2/2	0.6172 +/- 0.0110	1.0000 +/- 0.0000	0.3359 +/- 0.0110	0.8659 +/- 0.0067
s_t050_gibbs	2/2	0.5547 +/- 0.0773	1.0000 +/- 0.0000	0.3672 +/- 0.0110	0.8776 +/- 0.0045
r_t050_predictor	2/2	0.6562 +/- 0.0884	1.0000 +/- 0.0000	0.3672 +/- 0.0331	0.8743 +/- 0.0009
r_t050_gibbs	2/2	0.6172 +/- 0.0773	1.0000 +/- 0.0000	0.3203 +/- 0.0552	0.8769 +/- 0.0080

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
e_t050_predictor	1300	completed	64	64	0.5312	0.3281	9.11
e_t050_predictor	1301	completed	64	64	0.4062	0.2969	7.44
e_t050_gibbs	1300	completed	64	64	0.5625	0.2969	9.46
e_t050_gibbs	1301	completed	64	64	0.4688	0.2969	8.86
s_t050_predictor	1300	completed	64	64	0.5156	0.3281	9.25
s_t050_predictor	1301	completed	64	64	0.5781	0.3438	8.94
s_t050_gibbs	1300	completed	64	64	0.5625	0.3594	9.59
s_t050_gibbs	1301	completed	64	64	0.5781	0.3750	8.90

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
r_t050_predictor	1300	completed	64	64	0.5781	0.3906	9.24
r_t050_predictor	1301	completed	64	64	0.5000	0.3438	8.90
r_t050_gibbs	1300	completed	64	64	0.5000	0.3594	9.31
r_t050_gibbs	1301	completed	64	64	0.4531	0.2812	8.81

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: b50d97dcdd10f35c85aba61c9b9b948081ae9726809abb25b6c9e0a8834a92e9

v6-e-t050-predictor / seed 1300: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index":

"5"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-predictor/seed_1300/raw_samples.csv", "sha256": "e65e1f806e958601528d3c48b31e727f004318f427956ba7dd77ceece5d3b5d2", "size_bytes": 20407}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-e-t050-predictor / seed 1301: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 872}], "index": 6, "memory_total_mib": 49140, "memory_used_mib": 895, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "6"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-predictor/seed_1301/raw_samples.csv", "sha256": "30f4f1668bd4716728fe9fed2adbe5efc397a709c75716a17f57c8f0c57b167", "size_bytes": 20096}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-e-t050-gibbs / seed 1300: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fd6e3ce", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fd6e3ce"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-gibbs/seed_1300/raw_samples.csv", "sha256": "7912f9c622d82870edbb211c1688a73fb100b0cefd49e8a614a8622ae88db94b", "size_bytes": 19526}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-e-t050-gibbs / seed 1301: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0,

"softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-gibbs/seed_1301/raw_samples.csv", "sha256": "64c88ce12d8ac7a022d6b1715cbef0ffa2ec0434145bae6f9995895d94eb492f", "size_bytes": 20839, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-predictor / seed 1300: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fd6e6ce", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fd6e6ce"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-s-t050-predictor/seed_1300/raw_samples.csv", "sha256": "cfe608dd00991b6b95db83694542150cc55feeabfa262953d9a312f317ea3017", "size_bytes": 20527, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-predictor / seed 1301: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes":

{["pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}}, {"index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, {"launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, {"logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, {"physical_index": "5"}, {"raw": {"relative_path": "output/udlm/engineering_v6/v6-s-t050-predictor/seed_1301/raw_samples.csv", "sha256": "df3e72494039fd6a19e51489cfdea60f69d74bea55deee8f94c44b6d19c20ff0", "size_bytes": 20731}, {"sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, {"source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-gibbs / seed 1300: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, {"corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, {"model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, {"gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}}, {"index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, {"launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, {"logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, {"physical_index": "2"}, {"raw": {"relative_path": "output/udlm/engineering_v6/v6-s-t050-gibbs/seed_1300/raw_samples.csv", "sha256": "3d389848d51f373a2f911fb06c34048f23276cef913b5acb201495d40cfdb0d1", "size_bytes": 20711}, {"sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, {"source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-gibbs / seed 1301: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, {"corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, {"model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, {"gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}}, {"index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 6, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, {"launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, {"logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name":

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v6-r-t050-predictor / seed 1300: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": "5", "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-predictor/seed_1300/raw_samples.csv", "sha256": "72525be9b9e5ccfafa3c4af4a1310f56328499f291dfe5290d57789777f52d2", "size_bytes": 20472, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}}

v6-r-t050-predictor / seed 1301: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": "2", "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-predictor/seed_1301/raw_samples.csv", "sha256": "af6ecd7bbb3baad91217fe15a63e7fc88c2a06d1713b0cfe2ca71035ff6a22c2", "size_bytes": 21730, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}}

v6-r-t050-gibbs / seed 1300: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation

per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-gibbs/seed_1300/raw_samples.csv", "sha256": "82a3ed576ca875885098408c9279a0b75d1d33d4590b7d82bcb0ea23d8f32823", "size_bytes": 20943}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-r-t050-gibbs / seed 1301: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-gibbs/seed_1301/raw_samples.csv", "sha256": "a7f427c0741144239f64a594a7cca57bdbb98d8297cf62a2dbfb7065105dc678", "size_bytes": 20300}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

GenMol from-scratch benchmark

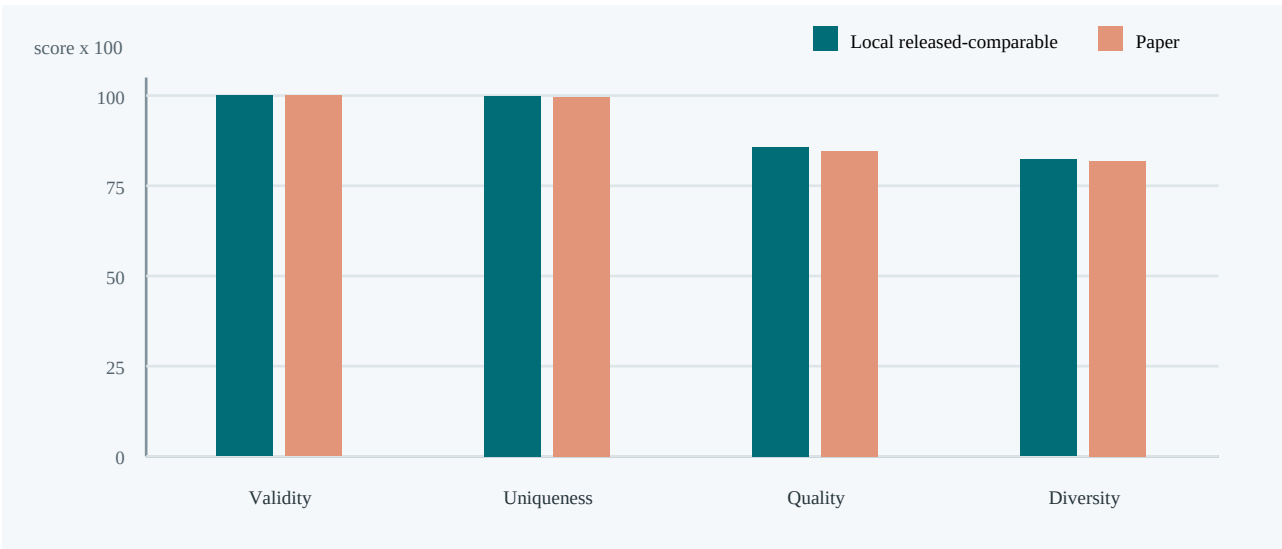
Final 50,000-step checkpoint compared with published GenMol V1 Table 1

COMPARATIVE RESULT - NOT AN EXACT REPRODUCTION CLAIM. All three local seeds completed with 1,000 retained raw rows and passed independent count, checkpoint, configuration, and artifact-hash validation.

Headline comparison

Metric	Local released-comparable	Published GenMol V1	Local - paper
Validity	100.00 +/- 0.00%	100.00 +/- 0.00%	+0.00 pp
Uniqueness	99.87 +/- 0.06%	99.70 +/- 0.10%	+0.17 pp
Quality	85.80 +/- 1.23%	84.60 +/- 0.80%	+1.20 pp
Diversity	0.8230 +/- 0.0013	0.8180 +/- 0.0010	+0.0050
Generation time	91.44 +/- 11.58 s	21.10 +/- 0.40 s	+70.34 s

Headline values are the arithmetic mean and sample SD of the three seed-level metrics. SD uses $\sqrt{\sum((x_i - \text{mean})^2)/(n-1)}$, $n=3$ (ddof=1); rows are not pooled before computing headline rates.



Bars show means on a 0-100 scale (diversity multiplied by 100). Uncertainty remains in the tables to avoid suggesting paired estimates.

Interpretation boundary

Differences are descriptive. The local checkpoint was trained from scratch with documented protocol deviations, and the published run seeds are undisclosed. No significance, parity, superiority, or exact reproduction conclusion follows from this table.

Seed-level results and decoding funnel

The released-comparable branch is the Table 1 comparison target. The strict branch exposes output that decodes without SAFE repair or largest-component selection.

Released-comparable metrics

Seed	Validity	Uniqueness	Quality	Diversity	Generation
0	100.00% (1000/1000)	99.80% (998/1000)	84.90% (849/1000)	0.8239 (n=998)	103.84 s
1	100.00% (1000/1000)	99.90% (999/1000)	87.20% (872/1000)	0.8215 (n=999)	80.92 s
2	100.00% (1000/1000)	99.90% (999/1000)	85.30% (853/1000)	0.8237 (n=999)	89.55 s
Mean +/- sample SD	100.00 +/- 0.00%	99.87 +/- 0.06%	85.80 +/- 1.23%	0.8230 +/- 0.0013	91.44 +/- 11.58 s

Strict metrics

Seed	Validity	Uniqueness	Quality	Diversity
0	99.00% (990/1000)	99.80% (988/990)	83.70% (837/1000)	0.8236 (n=988)
1	98.60% (986/1000)	99.90% (985/986)	86.10% (861/1000)	0.8206 (n=985)
2	98.80% (988/1000)	99.90% (987/988)	84.50% (845/1000)	0.8232 (n=987)
Mean +/- sample SD	98.80 +/- 0.20%	99.87 +/- 0.06%	84.77 +/- 1.22%	0.8225 +/- 0.0016

Strict vs repaired funnel (counts)

Seed	Requested / raw SAFE	Strict valid > unique > quality	Repair recovered / released lost	Released valid > unique > quality	Largest component applied
0	1000 / 1000	990 > 988 > 837	10 / 0	1000 > 998 > 849	14
1	1000 / 1000	986 > 985 > 861	14 / 0	1000 > 999 > 872	14
2	1000 / 1000	988 > 987 > 845	12 / 0	1000 > 999 > 853	13
Sum	3000 / 3000	2964 > 2960 > 2543	36 / 0	3000 > 2996 > 2574	41

Unique counts are summed within seed; molecules are not deduplicated across independent runs.

Exact metric definitions

Metric/path	Definition and denominator
Released Comparable	Released-code-compatible path: repair SAFE with fix=True, canonicalize, retain the largest disconnected component, then evaluate.
Strict	Strict path: canonical SAFE decode with fix=False and no largest-component selection. It is a diagnostic local addition and has no Table 1 counterpart.
Validity	valid_count / 1,000 requested samples, independently per seed.
Uniqueness	first-occurrence unique valid molecules / valid_count, per seed.
Quality	first-occurrence unique valid molecules with QED >= 0.6 and SA <= 4.0 / 1,000 requested samples, per seed.
Diversity	TDC Diversity on first-occurrence unique valid molecules; diversity_input_count records the input size.
Generation Time	Released-compatible wall-clock boundary: one 1,000-sample model/tokenizer call plus SAFE fix=True repair and largest-component selection. It equals model_sampling_and_tokenizer + released_postprocessing; model loading, strict decoding, and metric evaluation are excluded.

Checkpoint and protocol provenance

Checkpoint identity: global step 50,000; SHA-256
8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6. All three summaries agree.

Verified checkpoint

Field	Value
Resolved path	/home/aidar.alimbayev/Documents/genmolv2/outputs/paper_v1/checkpoints/50000.ckpt
SHA-256	8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6
Size	1,396,998,679 bytes
Global step	50,000
EMA	202 finite shadow tensors; 50,000 updates; decay 0.9999
Config SHA-256	42eadcb450cfc97f151157cae54d2a12ee88227d7368a9834b2fb75e40d40830
Effective-config SHA-256	8b6bcb62baff2b1837627ff1109fd3d20a73fc900e2d9b2dcb1f1b36b67213d
Sampling-config SHA-256	ee1c7815ed56d51b311ea4bab2c88100b5bdf2897c6ddb987d406e5b3096aec4
Runner SHA-256	a77d7c84d8628403c6d7a11edebce5bf9b8405e3f8b79a326a6e06bd7f444cea
Sampler source SHA-256	30b2e04282440471fb9059c35fa51ad4adcda9f7a81c838e2b5ca62073d1111f
Model source SHA-256	7e6004fbb0cc22f68988d76d52b0c6066194a8a2e7a182eaadbf73dd856402ab
Chemistry utilities SHA-256	0c7de4c7225f80e9a8cbbdb99018e5e8555c114ce6deea7dfcd593fc5e0f3867
Data utilities SHA-256	7d707bdef8edd6c5235935ca867ba7d799a00e0624a5ea9662665bfa2a94daa9
Bracket converter SHA-256	d3dca11fc12d7f4f0f439c03fc7d497f4efc6d87091388b7031c9d856279a618
Length distribution	n=249,455; min/median/max=10/49/87; SHA-256 9795cde72c60e9e58cd6dc99551801bdf47d00de6e060c8afb5f30584b19cf27

Audited evaluation protocol

Setting	Audited local value
Runs / seeds	3 independent invocations; explicit seeds 0, 1, 2
Samples	1,000 per seed; one generation batch; 3,000 requested total
Softmax temperature	0.5
Randomness	0.5
Minimum added length	40
Confidence reveal count	N=1, implicit MDLM.get_num_steps_confidence default used by released sampler
Released postprocessing	SAFE fix=True; canonical decode; retain component with maximum SMILES string length
Strict diagnostic	SAFE fix=False; no component selection
Generation timer	model_sampling_and_tokenizer + released_postprocessing

Loaded tokenizer provenance

Tokenizer field	Recorded inference value
Identity	requested=datamol-io/safe-gpt; runtime name=not retained; class transformers.tokenization_utils_fast.PreTrainedTokenizerFast
Revision	declared=not declared; resolved=not recorded
Vocabulary	base=1,880; effective=1,882 after repository-added tokens; SHA-256 8fdf74b22e1fa097c83b45022ddd1270adb45873962b0b5b3aba480e0ff8260f
Added/backend hashes	added=818312ee2fd79eb194e7c1e9677d671c46e8f6e13e0371cc6c01a4df196c241a; backend=not serializable
Special IDs	PAD=3, BOS=1, EOS=2, MASK=4

Documented deviations and published source

Known training and evaluation deviations

Item	Local	Published/reference	Why it matters
Training global batch size	2,046 (62 microbatch x 3 GPUs x 11 accumulation)	2,048	Small protocol deviation; checkpoint is not byte-identical to the released model.
Training hardware	3 NVIDIA RTX A6000 GPUs; 46:19:39 wall time	8 NVIDIA A100 GPUs; approximately 5 hours	Training and inference wall times are hardware-dependent and not speed reproductions.
Dataset and tokenizer revisions	datamol-io/safe-gpt identifiers were unpinned	Exact local revisions cannot be proven from this run	Possible corpus/tokenizer drift remains an uncontrolled reproducibility variable.
Evaluation seeds	Explicit seeds 0, 1, 2	Three runs; seed values undisclosed	Runs are repeatable locally but cannot be paired seed-for-seed with Table 1.
Training RNG seed	Not recorded in the Hydra training config or retained training log	Not disclosed	The training trajectory cannot be recreated seed-for-seed.

HARDWARE TIMING CAVEAT. Local generation time is measured on user-selected, shared RTX A6000 inference devices, using the released `de_novo_generation` timer boundary (model/tokenizer + SAFE repair + largest component). Each launch snapshot satisfied an exclusive utilization threshold strictly below 15% and the recorded free-memory minimum, but active compute processes were allowed. Table 1 used an A100 and a different software environment. Sharing can affect runtime, so the time delta is descriptive and is not a speed claim.

Published source

Primary paper PDF: <https://raw.githubusercontent.com/mlresearch/v267/main/assets/lee25o/lee25o.pdf>. Published values transcribed here: validity 100.0 +/- 0.0%, uniqueness 99.7 +/- 0.1%, quality 84.6 +/- 0.8%, diversity 0.818 +/- 0.001, and time 21.1 +/- 0.4 seconds.

The paper reports three 1,000-sample runs, but its seed values are undisclosed. Local seeds 0, 1, and 2 are therefore explicit repeatability choices, not matched replicas of paper runs.

Audit trail and claims boundary

Per-seed artifact identities

Seed	Raw CSV SHA-256	Summary SHA-256	Inference device
0	fb1a255f04092fdf8763b18ecb866eef02f1d7dde422727b00ce549bc7b2b2f	c8b312dd905ddd0f972920c2851569a4c2468c98aa46dfa2ac3f4dcbb3567a65	NVIDIA RTX A6000; physical 2, logical 0
1	5f20bd6731e6c095189de1f7c82ecd46123115b308b1b232c167d64afd1c6dbb	f95f8261b37f82a711848c18b7b15ebea1235858099b2bf8ef5d53b528699575	NVIDIA RTX A6000; physical 2, logical 0
2	70c962d46685855b8a1ac099e834956c32a1f0d8dda4a60146b9a8dbb1cac58a	a5c1be00950744b634c23ced6d2fea2f49ffe75b0d1419b3b01b9841bd3c4983	NVIDIA RTX A6000; physical 2, logical 0

Shared low-utilization GPU launch snapshots

Seed	User-selected physical ID	Point-in-time load / free memory	Active compute processes at selection
0	2 (allowed: 2)	5% (< 15%); 47,387 MiB free (minimum 30,000)	1: PID 1701406 /home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python (1730 MiB)
1	2 (allowed: 2)	5% (< 15%); 47,387 MiB free (minimum 30,000)	1: PID 1701406 /home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python (1730 MiB)
2	2 (allowed: 2)	5% (< 15%); 47,387 MiB free (minimum 30,000)	1: PID 1701406 /home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python (1730 MiB)

Validation performed before aggregation

Exactly seeds 0/1/2; completed summary markers; top-level/nested seed and sample-count agreement; 1,000 ordered CSV rows per seed; exact CSV schema; raw-file SHA-256; expected checkpoint hash/size/global step; exact V1 sampling/generation protocol; distinct ordered raw outputs; complete RNG seeding; user-selected physical-ID provenance and UUID to logical cuda:0 mapping; shared-GPU snapshots satisfying the recorded exclusive utilization and minimum-free-memory policy, with process inventories preserved; cross-seed dependency, tokenizer, implementation, checkpoint, effective-config, and runner identity; recomputed strict/released validity, first-occurrence uniqueness, quality gates, repair recovery, largest-component flags, failure counts, ratios, and timing invariants.

Caveats

1. This is a from-scratch implementation benchmark against published reference values, not a claim of exact model or environment reproduction.
2. The released-comparable path intentionally repairs malformed SAFE and may drop disconnected components. Strict results expose the effect; neither path should be silently substituted for the other.
3. With three local seeds, SD is descriptive. No hypothesis test, confidence interval, or equivalence claim is made. Paper seed values are unavailable.
4. Descriptive only: the local timer matches the released de_novo_generation boundary (model/tokenizer plus SAFE repair and largest-component selection), but local inference uses user-selected shared RTX A6000 hardware while the paper used an A100. Point-in-time selection below the recorded exclusive utilization threshold does not prevent concurrent workload interference, and the software environments are not identical.
5. Unpinned in the local training configuration; exact dataset and tokenizer revisions are unproven.
6. The local training RNG seed was not recorded. Evaluation seeds are explicit, but they do not repair that training-provenance gap.
7. Implementation hashes establish that all local seeds used the same code. This reporter does not assert that those hashes equal a pristine NVIDIA release checkout.

BOTTOM LINE. This document faithfully describes the completed local benchmark and its distance from the published headline values. It does not assert that the released checkpoint, original seeds, training data revision, hardware, or full environment were reproduced exactly.